Customer Churn Prediction Model Report (Task 2)

1. Introduction

Customer churn prediction is a critical application of data science in the banking industry. Retaining existing customers is significantly more cost-effective than acquiring new ones, making churn prediction an essential component of customer retention strategies.

The objective of this task is to develop a machine learning model capable of predicting whether a customer is likely to churn. By analysing historical customer data and behavioural patterns, the model aims to identify customers who may leave the bank in the future.

The insights generated from this predictive model will allow Lloyds Banking Group to proactively intervene with targeted retention strategies, thereby improving customer loyalty and long-term profitability.

2. Algorithm Selection

Selecting an appropriate machine learning algorithm is important for balancing predictive accuracy and interpretability. In this project, **Logistic Regression** was selected as the primary algorithm due to its suitability for binary classification problems and its interpretability.

Logistic Regression is commonly used in churn prediction because it provides a clear understanding of how different variables influence the likelihood of churn. This transparency is particularly valuable in a business environment where decision-makers must understand the reasoning behind model predictions.

In addition to Logistic Regression, **Random Forest** was considered as a comparative model. Random Forest is an ensemble learning algorithm capable of capturing more complex relationships in the data. Comparing these models helps determine whether a more sophisticated model significantly improves predictive performance.

3. Model Training Process

The dataset used for model training was the cleaned and preprocessed dataset produced in **Task 1**.

Several preprocessing steps were required to prepare the data for machine learning:

- Categorical variables were converted into a numerical format using **one-hot encoding**.
  Date fields were removed to avoid errors during model training.
- The target variable **ChurnStatus** was converted into binary form (0 = No Churn, 1 = Churn).

To evaluate model performance effectively, the dataset was divided into **training and testing sets** using an **80/20 split**. The training dataset was used to train the model, while the testing dataset was used to evaluate how well the model performs on unseen data.

Stratified sampling was applied during the split to ensure that the proportion of churned and non-churned customers remained consistent across both datasets.

Additionally, **class weighting** was used in the Logistic Regression model to address the imbalance between churned and non-churned customers.

4. Model Evaluation Metrics

Several evaluation metrics were used to assess the performance of the churn prediction model:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

These metrics provide a comprehensive evaluation of the model, especially in scenarios where the dataset is imbalanced.

Metric	Value
Accuracy	0.50
Precision	0.20
Recall	0.46
F1 Score	0.28
ROC-AUC	0.50

Fig 1: Logistic Regression Performance

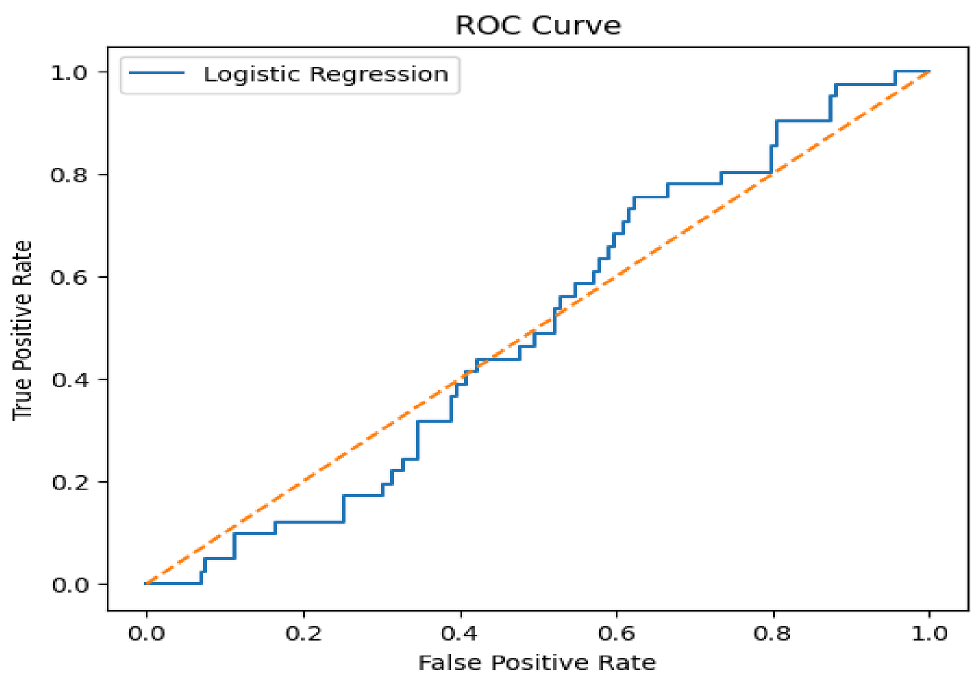

Fig 2: ROC curve demonstrating model discrimination capability

The **accuracy score of 0.50** indicates that the model correctly predicted 50% of the cases in the test dataset.

However, accuracy alone does not fully reflect model performance in churn prediction because the dataset is imbalanced.

The **precision score of 0.20** indicates that only 20% of customers predicted to churn actually churned.

The **recall score of 0.46** indicates that the model successfully identified 46% of the actual churned customers.

The **F1 score of 0.28** represents the balance between precision and recall.

The **ROC-AUC score of 0.50** suggests that the model's ability to distinguish between churn and non-churn cases is limited and close to random guessing.

Confusion Matrix Analysis

The confusion matrix produced by the model is shown below:

Actual / Predicted	No Churn	Churn
No Churn	81	78
Churn	22	19

Fig 3: Confusion matrix illustrating model prediction outcomes

The confusion matrix shows:

- 81 customers were correctly predicted as non-churn
- 19 customers were correctly predicted as churn
- 78 customers were incorrectly predicted as churn
- 22 churned customers were missed by the model

This indicates that the model struggles to distinguish between churned and non-churned customers, leading to a high number of incorrect predictions.

5. Model Comparison

Random Forest was also tested to evaluate whether a more complex model could improve predictive performance. Random Forest models are capable of capturing non-linear relationships between variables and often provide better performance in customer behaviour prediction tasks. The comparison between Logistic Regression and Random Forest helps determine whether interpretability or predictive strength should be prioritised.

6. Feature Importance and Insights

Feature importance analysis helps identify which variables most strongly influence customer churn.

Although Logistic Regression provides coefficient values rather than direct feature importance scores, these coefficients help determine which factors increase or decrease churn likelihood.

Key variables that may influence churn include:

- Frequency of customer transactions
- Customer service interactions
- Usage of digital banking services
- Customer demographic characteristics
- Income level and financial behaviour

Understanding these variables can provide valuable insights for designing effective customer retention strategies.

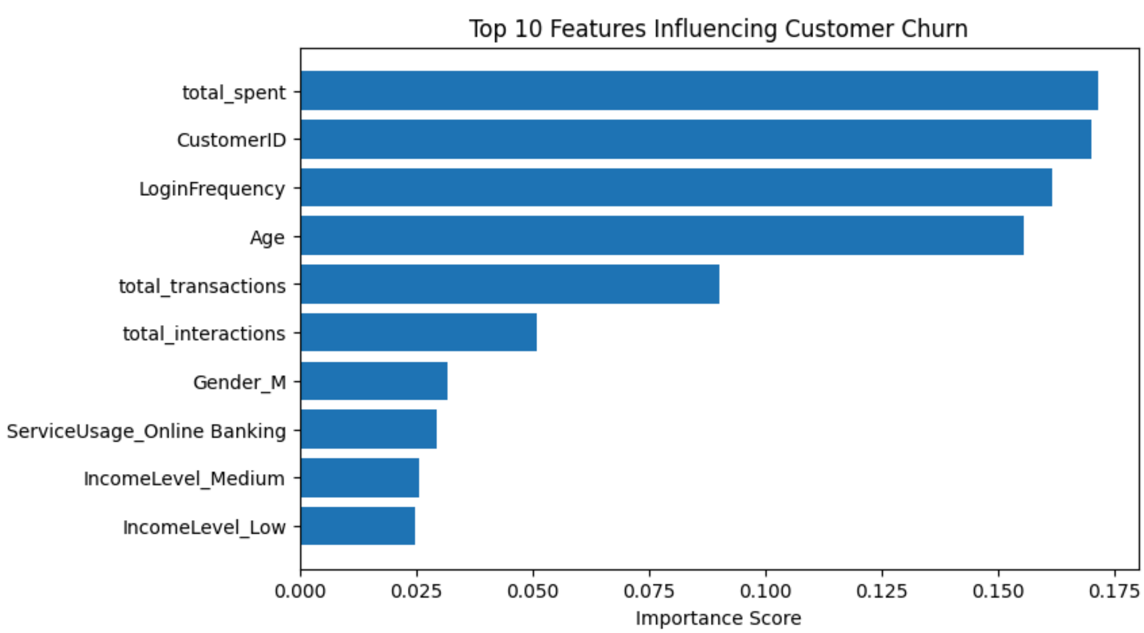

Fig 4: Top features influencing customer churn prediction

7. Business Application

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8. Potential Model Improvements

While the Logistic Regression model provides a useful baseline for churn prediction, several improvements could enhance its performance:

1. Implement more advanced machine learning algorithms such as **Random Forest, Gradient Boosting, or XGBoost**.

2. Apply **feature engineering** techniques such as calculating recency of transactions or login frequency.

3. Use **SMOTE or other resampling techniques** to address class imbalance.

4. Perform **hyperparameter tuning** to optimise model performance.

5. Continuously retrain the model using updated customer data to maintain predictive accuracy.

These improvements could significantly increase the model's ability to identify churn patterns.

9. Conclusion

This project demonstrates how machine learning techniques can be applied to customer data to predict churn and support strategic decision-making within Lloyds Banking Group.

Although the Logistic Regression model achieved moderate performance, it provides a valuable starting point for understanding customer behaviour and identifying churn risks.

With further refinement, additional features, and more advanced algorithms, predictive analytics can become a powerful tool for improving customer retention and enhancing business performance.

By leveraging data-driven insights, Lloyds Banking Group can proactively address churn risks and maintain a strong competitive position in the banking industry.